Programming Languages and Paradigms
Practice No. 2
Object Oriented Programming

Start date:	Oct. 14, 2026
Deliver date:	Stablished on EAFIT Interactiva
Delivery mechanism:	EAFIT Interactiva
Task value:	20% of the course

1 Objectives

- Apply basic concepts related to OO programming.
- Develop simple applications using C++.
- Present the development in a clear way.

2 Deliver and grading

2.1 To have into account ...

This practice have to be develop in couples mandatory, and in the extraordinary case someone is alone, a single group of tree people is allowed. The teacher is free to request an in-person support of the work to any student. This practice can be implemented in C++, auxiliary methods like matrix processing and statistical computing should be developed (do not use external libraries).

2.2 Deliver

What to put on EAFIT Interactiva?: Complete names of group students, repository link and YouTube video link in the dates allowed.

Report document should include:

- Student names, IDs end institutional emails.
- Documentation of every single method developed.
- Test evidence of the results on each item requested, with inputs and opurputs.

- Descriptions of troubles/solutions in the development and at least three conclusions.
- Video link.
- At leat 5 conclussions related to the job development.

Each couple should perform a video (up to 10 minutes) and publish it in YouTube (as not listed). In this video, the students say their names, explain the code and development strategies for implementation. The faces should be visible in the video and both students should participate.

2.3 Grading

Each student should be in capacity of answer any topic related to this practice. Grading scale goes from 0 to 50 points. The grading rubric will be:

- Correct functionability 40%.
- Practice presentation 40%.
- Code, commentaries and report 20%.

3 Activities to do

3.1 OOP Challenge

Design and implement an object-oriented framework to compute multiple linear regression models. The class have to implement, at least with:
Attributes:

- weights: Array with model weights.
- bias: The bias term.

Methods:

- fit(): Trains the model.
- predict(): Predicts outputs.
- score(): Computes predictions errors.
- data_scaling(): Scales the dataset values.

A pseudo code for test the models can look like this:

```
// Multiple linear regression
X_train[][] = ...; // Input data
y_train[] = ...; // Output data;
LinearRegression mlr = new LinearRegression(...);
mlr.fit(...);
print("weights: " + mlr.getWeigths());
print("bias: " + mlr.getBias());
X_test[][] = ...; // Test data
```

```
y_test[] = ...;
y_hat[] = mlr.predict(...);
for (yi in y_hat)
  print("y_hat: " + yi);
print("score: " + mlr.score(...));
```

3.1.1 Datasets

For train and prediction data, use the files available in EAFIT interactiva:

- For simple regression use *ice_cream selling_data.csv*.
- For multiple regression use *student_exam_scores.csv*.

3.2 Memory analysis

Create `LinearRegression` models in global, local and dinamic memory. Wich are the addresses of this objects? Wich are their sizes and why is that value? Explain clearly.

Create a vector of `LinearRegression` in dinamic memory, compute their size and justify the value.

Make a search of at least 3 historical catastrofical software fails due to dinamic memory management. For each one of them show:

- List the people/companies implied.
- What were the consecuences of the software fail.
- How that fail could be fixed.
- Link/reference to the consulted source.

3.3 Linear regression concepts

Basic linear regression concepts are covered in class sessions.

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